PRINCE SHRI VENKATESHWARA PADMAVATHY ENGINEERING COLLEGE (AUTONOMOUS)
MEDAVAKKAM - MAMBAKKAM MAIN ROAD, PONMAR, CHENNAI-127
APPROVED BY AICTE, NEW DELHI / AFFILIATED TO ANNA UNIVERSITY, CHENNAI
ACCREDITED WITH NAAC A++ / NIRF BAND : 201-300 /ISO 9001: 2015 CERTIFIED INSTITUTION
RECOGNIZED BY UGC UNDER SECTION 2(F) & 12(B) OF UGC ACT 1956

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

22EC29E-DIGITAL IMAGE PROCESSING

(B.E. ECE – V SEMESTER)

Regulation 2022

Academic Year (2025-2026)

Name of the Student:

Register Number :

Year / Semester :

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BONAFIDE CERTIFICATE

Name :

Register No :

Year/Semester :

Branch :

22EC29E–DIGITAL IMAGE PROCESSING LABORATORY

Certified that this is a Bonafide Record of the work done by the above student in the "**22EC29E–DIGITAL IMAGE PROCESSING LABORATORY"** in the academic year 2025-2026.

Submitted for Practical Examination held on ----------

Signature of Faculty In-Charge **Signature of Principal**

Internal Examiner **External Examiner**

VISION OF THE INSTITUTE

To be a prominent institution for technical education and research to meet the global challenges and demand for the societal needs.

MISSION OF THE INSTITUTE

- To develop the needed resources and infrastructure, and to establish a conducive ambience for the teaching- learning process.
- To nurture in the students, professional and ethical values, and to instill in them a spirit of innovation and entrepreneurship.
- To encourage in the students a desire for higher learning and research, to equip them to face the global challenges.
- To provide opportunities for students to get the needed additional skills to make them industry ready.
- To interact with industries and other organizations to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT

To become a Centre of excellence in the field of electronics and communications to produce a resourceful self-confident engineer to meet the global challenges and societal needs.

MISSION OF THE DEPARTMENT

M1. To imbue technical education through effective teaching learning process and state-of-art infrastructure.

M2. To foster the culture of innovation, research to impart comprehensive and multi-disciplinary projects to cater the global challenges.

M3. To imbibe professional ethics, leadership skills, social, cultural and environmental Awareness with passion for life- long learning.

PROGRAM OUTCOMES (POs)

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: Graduates will design, test and implement interdisciplinary and innovative systems in their profession through global education standards.

PEO 2: Graduates will inculcate effective communication skills, team work, professional ethics and preparing for leadership in a rapidly changing world.

PEO 3: Graduates will apply latest technologies of electronics and communication engineering for the current professional issues and come out with innovative solutions for the betterment of the nation and society.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1 To design and develop systems for real time problems in the areas related to electronics and communication engineering using latest Communication Systems, Embedded Systems and VLSI.

PSO 2 To develop innovative eco-friendly solutions and products in the field of electronics and communication engineering using cutting-edge hardware and software tools along with analytical skills.

INSTRUCTIONS TO STUDENTS

- Before entering the lab the student should carry the following things (MANDATORY)
    1. Identity card issued by the college.
    2. Class notes
    3. Lab observation book
    4. Lab Manual
    5. Lab Record
- Student must sign in and sign out in the register provided when attending the lab session without fail.
- Come to the laboratory in time. Students, who are late more than 15 min., will not be allowed to attend the lab.
- Students need to maintain 100% attendance in lab if not a strict action will be taken.
- All students must follow a Dress Code while in the laboratory
- Foods, drinks are NOT allowed.
- All bags must be left at the indicated place.
- Refer to the lab staff if you need any help in using the lab.
- Respect the laboratory and its other users.
- Workspace must be kept clean and tidy after experiment is completed.
- Read the Manual carefully before coming to the laboratory and be sure about what you are supposed to do.
- Do the experiments as per the instructions given in the manual.
- Copyall the programs to observation which are taught in class before attending the lab session.
- Students are not supposed to use floppy disks, pen drives without permission of lab- in charge.
- Lab records need to be submitted on or before the date of submission.

URSE OUTCOMES: On completion of the course, the students will be able to		Experiments List
CO1	Analyze the fundamental concepts of digital image processing.	1, 2
CO2	Apply spatial and frequency domain techniques on images.	3, 4, 5, 6, 7, 8
CO3	Analyze image segmentation and restoration techniques.	9, 10, 13
CO4	Evaluate various wavelet transforms and image compression techniques.	11, 12
CO5	Investigate various image representation and recognition techniques.	14, 15

CO-PO Matrices:

COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	1	1	-	-	1	2	1	-	1	1	1
CO2	3	3	2	1	1	-	-	1	2	1	-	1	1	1
CO3	3	2	2	1	1	-	-	1	2	1	-	1	1	1
CO4	3	3	1	1	1	-	-	1	2	1	-	1	2	2
CO5	3	2	2	1	1	-	-	1	2	1	-	1	2	2
AVG	**3**	**2.2**	**1.6**	**1**	**1**	-	-	**1**	**2.0**	**1**	-	**1**	**1.4**	**1.4**

1- Low, 2- Medium, 3-High, '-' No Correlation

Relevance of COs, POs and PSOs

Exp. No.	List of Experiments	Relevance of COs	Relevance of POs & PSOs
1	RGB to Grayscale transformation	CO1	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
2	Image sampling and quantization	CO1	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
3	Histogram processing and basic thresholding function	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
4	Intensity level transformation	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
5	DFT analysis of image	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
6	Analysis of Spatial and intensity resolution of images	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
7	Analysis of images with different color models	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
8	Implementation of image smoothening filters (Mean and Median of the image)	CO2	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
9	Image segmentation - Edge detection, line detection and point detection	CO3	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
10	Region based segmentation	CO3	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
11	Image compression techniques	CO4	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
12	Image restoration	CO4	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
13	Segmentation using watershed transformation	CO3	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
14	Boundary tracing code	CO5	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
15	Embedding Digital Watermarking	CO5	PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
Additional Experiments			
16	Morphological Processing		PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
17	Texture Analysis Using GLCM Code		PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
Open Ended Experiments			
18	Hybrid DCT-DWT Image Compression		PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2
19	Image Denoising and Enhancement Techniques		PO1, 2, 3, 4, 5, 8, 9, 10, 12, PSO1, 2

EX. NO	DATE	NAME OF THE EXPERIMENT	PAGE NO	MARKS	STAFF SIGN

EX.NO:1 DATE:	RGB TO GRAYSCALE TRANSFORMATION

AIM:

To write and execute program for RGB to grayscale transformation using software tool.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

1.RGB TO GRAYSCALE TRANSFORMATION

```
clc;
clear all;
 close all;
 I=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\plantsguru-flowering-
   plants-rose-red_.webp');
 subplot(2,2,1);
imshow(I);
title('original image');
 Im_gray=rgb2gray(I);
 subplot(2,2,2);
imshow(Im_gray);
title('gray scale');
```

VIVA QUESTIONS:

1. What is the difference between an RGB image and a grayscale image?

2. How can you convert an RGB image to grayscale in SCILAB?

3. Why are different weights assigned to R, G, and B channels when converting to grayscale?

4. What is the effect of converting an RGB image to grayscale on file size?

5. Why might you want to convert an image to grayscale before processing?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:2 DATE:	IMAGE SAMPLING AND QUANTIZATION

AIM:

To write and execute the program for image sampling and quantization using software tool.

APPARATUS REQUIRED:

2. Personal Computer
3. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) RESIZE

```
clc;
close all;
im=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\plantsguru-flowering-
   plants-rose-red_.webp');
subplot(3,3,1);
imshow(im);
title('original image');
im1=imresize(im,[1024 1024]);
im2=imresize(im,[1024 1024]/2);
im3=imresize(im,[1024 1024]/4);
im4=imresize(im,[1024 1024]/8);
im5=imresize(im,[1024 1024]/16);
im6=imresize(im,[1024 1024]/32);
subplot(3,3,2);
imshow(im1);
title('1024');
subplot(3,3,3);
imshow(im2);
title('512');
subplot(3,3,4);
imshow(im3);
title('256');
subplot(3,3,5);
imshow(im4);
title('128');
subplot(3,3,6);
imshow(im5);
title('64');
subplot(3,3,7);
imshow(im6);
title('32');
```

B) CROPPING

```
clc;
i=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\plantsguru-flowering-
   plants-rose-red_.webp');
subplot(2,2,1);
imshow(i);
title('original image');
j=imcrop(i,[50,60,60,80]);
subplot(2,2,2);
imshow(j);
title('cropped image');
```

C) ZOOMING

```
clc;
clear all;
close all;
i=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\plantsguru-flowering-
   plants-rose-red_.webp');
subplot(2,2,1);
imshow(i);
title('original image');
roi=[10,20,40,70];
zoomed_I=imcrop(i,roi);
subplot(2,2,2);
imshow(zoomed_I);
title('zoomed image');
```

D) ROTATING

```
clc;
clear all;
close all;
a=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\plantsguru-flowering-
   plants-rose-red_.webp');

subplot(2,2,1);
imshow(a);
title('original image');

rot30=imrotate(a,30);
subplot(2,2,2);
imshow(rot30);
title('30 degree rotated image');

rot60=imrotate(a,60);
subplot(2,2,3);
imshow(rot60);
title('60 degree rotated image');

rot90=imrotate(a,90);
subplot(2,2,4);
imshow(rot90);
title('90 degree rotated image');
```

VIVA QUESTIONS:

1. What is sampling in the context of image processing?

2. What is the effect of sampling on an image?

3. What is the relationship between the number of quantization levels and image quality?

4. What is quantization in image processing?

5. What is the effect of quantization on an image?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:3 DATE:	HISTOGRAM PROCESSING AND BASIC THRESHOLDING FUNCTION

AIM:

To write and execute the program for Histogram processing and thresholding function.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) HISTOGRAM

```
clc;
A = imread('C:\Users\SWETHA\Downloads\butterfly.jpeg');
I = rgb2gray(A);

// Original image
subplot(2,2,1);
imshow(I);
title("Original Image");

// Histogram before equalization
subplot(2,2,2);
[counts, cells] = imhist(I);
bar(cells, counts);
title("Before Equalization");

// Histogram equalization
Ieq = imhistequal(I);

// Equalized image
subplot(2,2,3);
imshow(Ieq);
title("Equalized Image");

    // Histogram after equalization

subplot(2,2,4);
[counts2, cells2] = imhist(Ieq);
bar(cells2, counts2);
```

```
title("After Equalization");
```

B) THRESHOLD

```
// Read image
I = imread('C:\Users\SWETHA\Downloads\butterfly.jpeg');
I = rgb2gray(I); // convert to grayscale if needed

// Display original image
figure();
imshow(I);
title("Original Image");

// Normalize image to [0,1] for Otsu threshold
I_norm = double(I) / 255;

// Compute threshold using Otsu
level = imgraythresh(I_norm);

// Apply threshold
BW = I_norm > level;

// Display binary segmented image
figure();
imshow(BW);
title("Otsu Thresholded Image");
```

VIVA QUESTIONS:

1. What is a histogram in the context of image processing?

2. How can you calculate and display the histogram of an image in SCILAB?

3. Why is histogram processing important in image processing?

4. What is histogram equalization?

5. What effect does histogram equalization have on an image?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:4 DATE:	INTENSITY LEVEL TRANSFORMATION

AIM:

To write and execute the program for intensity level transformation.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) CONTRAST STRECHING

```
clc;
clear all;
close all;
c=imread('C:\Users\SWETHA\OneDrive\Pictures\Screenshots\lotus.jpeg');
c1=rgb2gray(c);
I2=im2double(c1);
m=mean2(I2);
eps=2.2204e-16;
contrast1=1./(1+(m./(I2+eps)).^4);
contrast2=1./(1+(m./(I2+eps)).^5);
contrast3=1./(1+(m./(I2+eps)).^-1);
subplot(2,2,1);
imshow(c);
title('ORIGINAL IMAGE');
subplot(2,2,2);
imshow(contrast1);
title('CONTRAST STRECHING(E=0.4)');
subplot(2,2,3);
imshow(contrast2);
title('CONTRAST STRECHING(E=0.5)');
subplot(2,2,4);
imshow(contrast3);
title('CONTRAST STRECHING(E=-1)');
```

B) INTENSITY LEVEL SLICING

```
clc;
clear all;
close all;
I=imread('C:\Users\SWETHA\Downloads\images.jpeg');
image=I(:,:,1);
```

```
rmin=100;
rmax=180;
[r,c]=size(image);
s=zeros(r,c);
for i=1:r
    for j=1:c
        if(rmin<image(i,j)&&image(i,j)<rmax)
           s(i,j)=0;
        else
           s(i,j)=image(i,j);
        end
    end
end
subplot(2,2,1);
imshow(I);
title('original image');
subplot(2,2,2);
imshow(s);
title('intensity level sicing');
```

C) LOG TRANSFORMATION

```
clc;
clc;
clear all;
close all;
I=imread('C:\Users\SWETHA\Downloads\images.jpeg');
J=255-I;
subplot(3,2,1);
imshow(I);
title('original image');
subplot(3,2,2);
imshow(J);
title('negative transformation');
a=rgb2gray(I);
I2=im2double(a);
K1=1*log(1+I2);
K2=3*log(1+I2);
K3=5*log(1+I2);
subplot(3,2,3);
imshow(K1);
title('log transformation(c=1)');
subplot(3,2,4);
imshow(K2);
title('log transformation(c=3)');
subplot(3,2,5);
imshow(K3);
title('log transformation(c=5)');
```

D)TRANSFORMATION

```
clc;
clear all;
close all;
I=imread('C:\Users\SWETHA\Downloads\images.jpeg');
//imshow(I);
J=255-I;
subplot(3,2,1);
imshow(I);
title('original image');
subplot(3,2,2);
imshow(J);
title('negative transformation');
a=rgb2gray(I);
I2=im2double(a);
K1=1*I2.^1;
K2=3*I2.^2;
K3=5*I2.^3;
subplot(3,2,3);
imshow(K1);
title('gamma transformation(c=1)');
subplot(3,2,4);
imshow(K2);
title('gamma transformation(c=3)');
subplot(3,2,5);
imshow(K3);
title('gamma transformation(c=5)');
```

VIVA QUESTIONS:

1. What is intensity level transformation in image processing?

2. Explain the logarithmic transformation and provide SCILAB code for it.

3. What is the power-law (gamma) transformation, and how would you implement it in SCILAB?

4. How does changing the gamma value affect the image?

5. What is the difference between linear and nonlinear intensity transformations?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:5 DATE:	DFT ANALYSIS OF IMAGE

AIM:

To write and execute the program for DFT analysis of the image.

APPARATUS REQUIRED:

1. Personal Computer
2. Softtware Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

DFT ANALYSIS OF AN IMAGE

```
Clc;
Clear all;
close all;
A=imread('C:\Users\SWETHA\Downloads\images.jpeg');
subplot(3,3,1);
imshow(A);
a=rgb2gray(A);
I2=im2double(a);
title('given image');
y=fft2(I2);
subplot(3,3,2);
imshow(y);
title('fft of the given image');
z=fftshift(y);
subplot(3,3,3);
imshow(z);
title('fft with shifted origin');
m=ifftshift(z);
subplot(3,3,4);
imshow(m);
title('fft reshifted to the original position');
n1=uint8(m);
subplot(3,3,5);
imshow(n1);
title('ifft of the given image');
b=dct(I2);
subplot(3,3,6);
```

```
imshow(b);
title('dct of the given image');
c=idct(b);
subplot(3,3,7);
imshow(c);
title('idct of the given image');
```

VIVA QUESTIONS:

1. What is the purpose of the Discrete Fourier Transform(DFT) in image processing?

2. How do you compute the DFT of an image in SCILAB?

3. Explain the concept of frequency domain filtering. Why is it sometimes more efficient than spatial filtering?

4. What information can you gather from the magnitude and phase of the DFT?

5. What is the effect of changing image size on the frequency content representation in the DFT?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:6 DATE:	ANALYSIS OF SPATIAL AND INTENSITY RESOLUTION OF IMAGES

AIM:

To write and execute the program for the analysis of spatial and intensity Resolution of images.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) SPATIAL RESOLUTION

```
clc;
clear;
close;
A = imread('C:\Users\SWETHA\Downloads\bts dip.jpeg');
C = rgb2gray(A);
C = im2double(C);
levels = 8;
nImage = round(C * (levels-1)) / (levels-1);
subplot(2,2,1);
imshow(C);
title("Original Grayscale Image");
subplot(2,2,2);
imshow(nImage);
title("change in spatial Resolution");
whos A
whos C
whos ('nImage');
```

B) INTENSITY RESOLUTION

```
I=imread('C:\Users\SWETHA\Downloads\park dip.jpeg');
[r,c]=size(I);
I2=uint8(zeros(r,c));
  for i=1:r for j=1:c
    if(I(i,j)>128)
      I2(i,j)=255;
    else
       I2(i,j)=0;
    end
   end
```

```
 end
figure;
subplot(2,2,1);
imshow(I);
title('original image');
subplot(2,2,2);
imshow(I2);
title('change in intensity resolution');
```

VIVA QUESTIONS:

1. What is spatial resolution in image processing?

2. What is the effect of reducing spatial resolution on an image?

3. How does spatial resolution relate to image quality?

4. What is intensity resolution in image processing?

5. What is the effect of decreasing intensity resolution on an image?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:7 DATE:	ANALYSIS OF IMAGES WITH DIFFERENT COLOR MODELS

AIM:

To write and execute the program for the analysis of images with different color models.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

COLOR MODELS

```
clc;
clear all;
close all;
img=imread('C:\Users\SWETHA\Downloads\pikachu.jpeg');
hsv_img= rgb2hsv(img);
cmy_img=imcomplement(img);
subplot(2,2,1);
imshow(img);
title('ORIGINALIMAGE');
subplot(2,2,2);
imshow(img);
title('RGB ');
subplot(2,2,3);
imshow(hsv_img);
title('HSI');
subplot(2,2,4);
imshow(cmy_img);
title('CMY');
```

VIVA QUESTIONS:

1. What is a color model, and why is it used in image processing?

2. What is the RGB color model?

3. What is the advantage of using the HSV model over RGB for image segmentation?

4. How can you visualize the three channels of an RGB image separately in SCILAB?

5. Explain a scenario where you would prefer using the YCbCr model over RGB.

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:8 DATE:	IMPLEMENTATION OF IMAGE SMOOTHENING FILTERS (MEAN AND MEDIAN OF THE IMAGE)

AIM:

To write and execute the program for the implementation of image smoothening filters.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

SMOOTHENING AND SHARPENING FILTERS

```
Clc;
clear all;
close all;
img=imread("C:\Users\SWETHA\Downloads\dip image.jpeg");
gray_img=rgb2gray(img);
f=fspecial('average',[1*1]);
mean1=filter2(gray_img,f);
mean1=uint8(mean1);
med=median(gray_img,[3*3]);
figure;
subplot(1,3,1);
imshow(img);
title('Original image');
subplot(1,3,2);
imshow(mean1);
title('Mean filtered image');
subplot(1,3,3);
imshow(uint8(med));
title('Median filtered image');
```

VIVA QUESTIONS:

1. What is image smoothing in image processing?

2. Why do we use smoothing filters in image processing?

3. What is a mean filter, and how does it work?

4. What is a Gaussian filter, and how does it differ from a mean filter?

5. What is a median filter, and how does it work?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:9 DATE:	IMAGE SEGMENTATION - EDGE DETECTION, LINE DETECTION AND POINT DETECTION

AIM:

To write and execute the program for different methods of image segmentation.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) EDGE DETECTION

```
clc;
clear;
x = imread("C:\Users\SWETHA\Downloads\dip image.jpeg");
subplot(2,3,1);
imshow(x);
title("Given image for edge detection");
// Convert to grayscale
   x = rgb2gray(x);

x = im2double(x); // normalize image
// Prewitt
b = edge(x, "prewitt");
subplot(2,3,2);
imshow(b);
title("Prewitt Edge Detection");
// Sobel
c = edge(x, "sobel");
subplot(2,3,3);
imshow(c);
title("Sobel Edge Detection");
// Canny
d = edge(x, "canny");
subplot(2,3,4);
imshow(d);
```

```
title("Canny Edge Detection");
// Roberts (manual implementation)
roberts_x = [1 0; 0 -1];
roberts_y = [0 1; -1 0];
gx = conv2(x, roberts_x, "same");
gy = conv2(x, roberts_y, "same");
roberts_edge = sqrt(gx.^2 + gy.^2);
roberts_edge = roberts_edge > 0.1; // threshold
subplot(2,3,5);
imshow(roberts_edge);
title("Roberts Edge Detection");
// Laplacian of Gaussian (LoG)
f = edge(x, "log");
subplot(2,3,6);
imshow(f);
title("LOG Edge Detection");
```

B) LINE DETECTION

```
clc;
clear;
close;
s = imread("C:\Users\SWETHA\Downloads\seouldip.jpg");
// Convert to grayscale manually
R = double(s(:,:,1));
G = double(s(:,:,2));
B = double(s(:,:,3));
f = 0.2989*R + 0.5870*G + 0.1140*B;
// -45 degree edge detector mask
w = [2 -1 -1; -1 -2 -1; -1 -1 2];
// Apply filter using convolution
gl = conv2(f, w, "same");
// Zoomed top-left region
gtop = gl(1:120, 1:120);
ghop = imresize(gtop, 4);
// Zoomed bottom-right region
[r, c] = size(gl);
gbot = gl(r-119:r, c-119:c);
gbot = imresize(gbot, 4);
// Thresholding
g = abs(gl);
T = max(g(:));
g_bin = g >= T;
// Plot results
scf(0);
```

```
subplot(3,3,1);
imshow(uint8(s));
xtitle("original image");
subplot(3,3,2);
imshow(uint8(f));
xtitle("grayscale image");
subplot(3,3,3);
imshow(mat2gray(gl));
xtitle("Result of processing with -45 degree detector");
subplot(3,3,4);
imshow(mat2gray(ghop));
xtitle("Zoomed view of the top left region of gl");
subplot(3,3,5);
imshow(mat2gray(gbot));
xtitle("Zoomed view of the bottom right region of gl");
subplot(3,3,6);
imshow(g_bin*255);
xtitle("Thresholded Image");
subplot(3,3,7);
imshow(g_bin*255);
xtitle("Final Result");
```

C) POINT DETECTION

```
clc;
clear;
close;
f=imread("C:\Users\SWETHA\Downloads\seouldip.jpg");
if size(f, 3) == 3
   f=rgb2gray(f);
end
w=[-1,-1,-1;-1,8,-1;-1,-1,-1];
g=abs(imfilter(double(f),w));
T = max(g(:));
g=g>=T;
subplot(1,2,1);
imshow(f);
title('Original Image');
subplot(1, 2, 2);
imshow(g);
title('Result of Point Detection');
```

VIVA QUESTIONS:

1. What is image segmentation in image processing?

2. Why is image segmentation important?

3. What is edge detection, and why is it important?

4. What are the differences between edge, line, and point detection?

5. What are some common edge detection methods?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:10 DATE:	REGION BASED SEGMENTATION

AIM:

To write and execute the program for region growing, splitting and merging.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) REGION GROWING

```
clc;
clear;
close;
I = imread("C:\Users\SWETHA\Downloads\winnie.jpeg");
I = im2double(rgb2gray(I));
[rows, cols] = size(I);
seed_x = round(rows / 2);
seed_y = round(cols / 2);
T = 0.08;
region = zeros(rows, cols);
region(seed_x, seed_y) = 1;
neighbors = [-1 -1; -1 0; -1 1;
             0 -1;       0 1;
             1 -1;  1 0;  1 1];

region_mean = I(seed_x, seed_y);
region_size = 1;
change =1;
while change > 0
   change = 0;
   [r_idx, c_idx] = find(region == 1);
   for k = 1:length(r_idx)
      for n = 1:8
         x = r_idx(k) + neighbors(n, 1);
         y = c_idx(k) + neighbors(n, 2);
```

```
if (x > 0 & x <= rows & y > 0 & y <= cols)
            if region(x, y) == 0 then
               diff = abs(I(x, y) - region_mean);
               if diff < T then
                  region(x, y) = 1;
                  region_mean = (region_mean * region_size + I(x, y)) / (region_size + 1);
                  region_size = region_size + 1;
                  change = change + 1;
               end
            end
         end
      end
   end
end
figure();
subplot(2,2,1);
imshow(I);
title("Original Image");

subplot(2,2,2);
imshow(region);
title("Segmented Image (Region Growing)");
```

B) REGION SPLITTING AND MERGING

```
clc;
clear;
close;
I = imread("C:\Users\SWETHA\Downloads\winnie.jpeg");  // change path
if size(I,3) == 3 then
   I = rgb2gray(I);
end
I = im2double(I);
[m, n] = size(I);
T = 0.01;      // Threshold for variance (lower = more boxes)
min_size = 4;   // Minimum region size
global output;
output = zeros(m, n);  // start with pure black image
function split_region(img, x, y, w, h)
   global output;
   sub_img = img(x:x+w-1, y:y+h-1);
   std_val = stdev(sub_img);

   if std_val > T & w > min_size & h > min_size then
      half_w = floor(w / 2);
```

```
        half_h = floor(h / 2);
        split_region(img, x, y, half_w, half_h);
        split_region(img, x, y + half_h, half_w, h - half_h);
        split_region(img, x + half_w, y, w - half_w, half_h);
        split_region(img, x + half_w, y + half_h, w - half_w, h - half_h);
    else
        // Draw white boundary box
        x1 = x; y1 = y; x2 = x + w - 1; y2 = y + h - 1;
        output(x1, y1:y2) = 1;    // top
        output(x2, y1:y2) = 1;    // bottom
        output(x1:x2, y1) = 1;    // left
        output(x1:x2, y2) = 1;    // right
    end
endfunction
split_region(I, 1, 1, m, n);
scf(0);
subplot(1,2,1);
imshow(I);
title("Original Image");

subplot(1,2,2);
imshow(output);
title("Output Image - Region Splitting and Merging");

disp("Region Splitting and Merging Completed!");
```

VIVA QUESTIONS:

1. What is region growing in image segmentation?

2. What is region splitting and merging in segmentation?

3. Why are region-based segmentation methods used?

4. How can you choose a good seed point for region growing?

5. What are common challenges associated with region-based segmentation methods?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:11 DATE:	IMAGE COMPRESSION TECHNIQUES

AIM:

To write and execute the program for lossy and lossless compression.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

A) LOSSY COMPRESSION

```
clc;

clear;

close;

input_image1 = imread("C:\Users\SWETHA\Downloads\nature dip.jpeg"); // Change this path

input_image1 = im2double(input_image1);

dims = size(input_image1);

if length(dims) == 3 then

   input_image1 = 0.2989*input_image1(:,:,1) + 0.5870*input_image1(:,:,2) +
0.1140*input_image1(:,:,3);

end


function [cA, cH, cV, cD]=haar2D(I)

   [rows, cols] = size(I);

   rows = rows - modulo(rows,2);

   cols = cols - modulo(cols,2);

   I = I(1:rows, 1:cols);
```

```
    cA = (I(1:2:rows,1:2:cols) + I(1:2:rows,2:2:cols) + I(2:2:rows,1:2:cols) +
I(2:2:rows,2:2:cols)) / 4;

    cH = (I(1:2:rows,1:2:cols) - I(1:2:rows,2:2:cols) + I(2:2:rows,1:2:cols) - I(2:2:rows,2:2:cols))
/ 4;

    cV = (I(1:2:rows,1:2:cols) + I(1:2:rows,2:2:cols) - I(2:2:rows,1:2:cols) - I(2:2:rows,2:2:cols))
/ 4;

    cD = (I(1:2:rows,1:2:cols) - I(1:2:rows,2:2:cols) - I(2:2:rows,1:2:cols) + I(2:2:rows,2:2:cols))
/ 4;

endfunction

[cA, cH, cV, cD] = haar2D(input_image1);

[rows, cols] = size(cA);

transformed = zeros(rows*2, cols*2);

transformed(1:rows, 1:cols) = cA;

transformed(1:rows, cols+1:$) = cH;

transformed(rows+1:$, 1:cols) = cV;

transformed(rows+1:$, cols+1:$) = cD;

transformed = transformed - min(transformed);

transformed = transformed / max(transformed);

figure();

subplot(1,2,1);

imshow(input_image1);

title("Original image");

subplot(1,2,2);

imshow(transformed);

title("Transformation image");
```

B) LOSSLESS COMPRESSION

```
clc;
clear all;
close all;
I = imread("C:\Users\SWETHA\Downloads\nature dip.jpeg");     // change image name if
   needed
if size(I,3) == 3 then
```

```
    I = rgb2gray(I);        // convert to grayscale if color
end
figure;
imshow(I);
title("Original Image");
[r, c] = size(I);
Ivec = I(:);
pixel = Ivec(1);
count = 1;
compressed = [];
for i = 2:length(Ivec)
    if Ivec(i) == pixel then
        count = count + 1;
    else
        compressed = [compressed; pixel, count];
        pixel = Ivec(i);
        count = 1;
    end
end
compressed = [compressed; pixel, count]; // last value

disp("Original Pixels: " + string(length(Ivec)));
disp("Compressed Pairs: " + string(size(compressed,1)));
disp("Compression Ratio: " + string(length(Ivec) / size(compressed,1)));

                                        decompressed = [];
for i = 1:size(compressed,1)
    decompressed = [decompressed; ones(compressed(i,2),1)*compressed(i,1)];
end

I_restored = matrix(decompressed, r, c);

figure;
imshow(uint8(I_restored));
title("Decompressed (Restored) Image");

if sum(abs(I(:) - I_restored(:))) == 0 then
    disp("Lossless Compression Achieved!");
else
    disp("Images differ - check steps.");
end
```

VIVA QUESTIONS:

1. What is image compression?

2. Why is image compression important?

3. What are the two main types of image compression?

4. What is the difference between lossy and lossless compression?

5. What is a compression ratio, and why is it important?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:12 DATE:	IMAGE RESTORATION

AIM:

To write and execute the program for image restoration techniques.

APPARATUS REQUIRED:

1. Personal Computer
2. Sofware Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

IMAGE RESTORATION

```
clc;
clear;
close();
I = imread("C:\Users\SWETHA\Downloads\leaf dip.jpeg");
I = im2double(I);
// Convert to grayscale if RGB
if size(I,3) == 3 then
   I = rgb2gray(I);
end
figure(1);
imshow(I);
title("Original Image");
len = 21;              // blur length
PSF = zeros(1, len);
PSF(:) = 1 / len;      // simple 1-D horizontal motion blur
blurred = conv2(I, PSF, "same");
```

```
figure(2);
imshow(blurred);
title("Blurred Image");
noise_var = 0.001;
noise = sqrt(noise_var) * rand(size(I,1), size(I,2));
blurred_noisy = blurred + noise;
figure(3);
imshow(blurred_noisy);
title("Simulate Blur and Noise");
[M, N] = size(I);
h = zeros(M, N);
h(1,1:len) = PSF;
H = fft2(h);
G = fft2(blurred_noisy);

K = 0.001;
W = (conj(H)) ./ ((abs(H).^2) + K);
F_hat = G .* W;
restored = real(ifft2(F_hat));

restored = restored - min(restored);
restored = restored / max(restored);
figure(4);
imshow(restored);
title("Restored Image (Wiener Filter)");
```

VIVA QUESTIONS:

1. What is image restoration?

2. What are some common causes of image degradation?

3. What is the difference between image restoration and image enhancement?

4. What are lossy and lossless image restoration techniques?

5. What is the difference between image restoration and image compression?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:13 DATE:	SEGMENTATION USING WATERSHED TRANSFORMATION

AIM:

To write and execute the program for segmentation technique using watershed transformation.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

SEGMENTATION USING WATERSHED TRANSFORMATION

```
clc;
clear;
close;
I = imread("C:\Users\SWETHA\Downloads\seouldip.jpg");
if size(I,3) == 3 then
   I = rgb2gray(I);
end
se = imcreatese("ellipse", 10, 10);
I1 = imtophat(I, se);
I2 = imadjust(I1);
threshold = mean(double(I2(:)));
BW = I2 > threshold;
C = ~BW;
[r, c] = size(C);
distMap = zeros(r, c);
```

```
D = C;


for i = 1:15
    D = imdilate(D, se);
    D = D(1:r, 1:c);   // ensure same size
    distMap = distMap + D;
end
distMap = distMap / max(distMap(:));
mask = distMap > 0.5;
seg = I2;
seg(~mask) = 0;
figure();
subplot(2,2,1);
imshow(I);
title("Original Image");
subplot(2,2,2);
imshow(I1);
title("Top-hat");
subplot(2,2,3);
imshow(I2);
title("Contrast Adjusted");
subplot(2,2,4);
imshow(seg);
title("Segmented Image");
```

VIVA QUESTIONS:

1. What is image segmentation, and why is it important?

2. Explain the watershed transformation in image segmentation.

3. What are the advantages of using the watershed transformation for segmentation?

4. What are some limitations of the watershed algorithm?

5. How does SCILAB implement the watershed algorithm?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:14 DATE:	BOUNDARY TRACKING CODE

AIM:

To write and execute the program for boundary tracking code**.**

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

BOUNDARY TRACKING

```
clc;
clear;
close;
I = imread("C:/Users/s.pooja26/Downloads/OIP (4).jpg"); // Use forward slashes
if ndims(I) == 3 then
    I = rgb2gray(I);
end
I = im2double(I);
threshold = 0.5;          // Adjust if needed
binaryImage = I > threshold;
figure();
imshow(binaryImage);
title("Boundary Tracking");
hold on;
                    [rows, cols] = size(binaryImage);
```

```
for i = 2:rows-1
    for j = 2:cols-1
        if binaryImage(i,j) == 1 then
            // Check if any neighbor is 0 (edge pixel)
            if sum(binaryImage(i-1:i+1, j-1:j+1) == 0) > 0 then
                plot(j, i, 'r.', 'MarkerSize', 1); // Plot boundary pixel
            end
        end
    end
end

hold off;
```

VIVA QUESTIONS:

1. What is boundary tracing in image processing?

2. Why is boundary tracing important in image analysis?

3. What types of images are commonly used for boundary tracing?

4. What are connected components in the context of boundary tracing?

5. What is the purpose of using the bw boundaries function in SCILAB?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:15 DATE:	EMBEDDING DIGITAL WATERMARKING

AIM:

To write and execute the program for embedding digital watermarking.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

EMBEDDING DIGITAL WATERMARKING

```
clc;

clear all;

close all;

// Read the host image

host = imread("C:\Users\SWETHA\Downloads\leaf dip.jpeg");

host = imresize(host, [256 256]);

// Read the watermark image

watermark = imread("C:\Users\SWETHA\Downloads\winnie.jpeg");

watermark = imresize(watermark, [256 256]);

// Convert both to grayscale if needed

if size(host,3) == 3 then

   host = rgb2gray(host);

end

if size(watermark,3) == 3 then

   watermark = rgb2gray(watermark);

end
```

```
// Display original images
figure();
subplot(1,2,1);
imshow(host);
title('Original Host Image');

subplot(1,2,2);
imshow(watermark);
title('Original Watermark Image');

// Normalize images
host = double(host);
watermark = double(watermark);

// --- EMBEDDING PROCESS ---
watermarked = bitand(host, 254) + bitand(watermark, 1); // Replace LSB of host with watermark LSB
figure();
imshow(uint8(watermarked));
title('Watermarked Image');

// --- EXTRACTION PROCESS ---
extracted = bitand(watermarked, 1) * 255;

// Display extracted watermark
figure();
imshow(uint8(extracted));
title('Extracted Watermark Image');
```

VIVA QUESTIONS:

1. What is digital watermarking?

2. What are the types of digital watermarking?

3. What are the primary applications of digital watermarking?

4. What is the process of embedding a watermark in digital content?

5. What are the commonly used algorithms for digital watermarking?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:16 DATE:	MORPHOLOGICAL PROCESSING

AIM:

To write and execute the program for morphological processing

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tool.

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

BASIC MORPHOLOGICAL OPERATION

```
I = imread('C:\Users\kaviya\Downloads\football.jpeg');
subplot(2,3,1);
 imshow(I);
title('Original image');
se= strel('line',7,7);
dilate = imdilate(I,se);
subplot(2,3,2);
imshow(dilate);
 title('Dilate image');
erode = imerode(I,se);
subplot(2,3,3);
 imshow(erode);
title('Erode image');
open= imopen(I,se);
subplot(2,3,4);
imshow(open);
title('Open image');
```

```
closed = imclose(I,se);
subplot(2,3,5);
imshow(closed);
title('Closed image');
```

VIVA QUESTIONS:

1.What is morphological processing in image analysis?

2. What is dilation, and how does it work?

3. What is erosion, and what is its effect on an image?

4.What is erosion, and what is its effect on an image?

5.What is the closing operation, and when would you use it?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:17 DATE:	TEXTURE ANALYSIS USING GLCM CODE

AIM:

To write and execute the program for texture analysis using GLCM code.

APPARATUS REQUIRED:

1. Personal Computer
2. Software Tools

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

TEXTURE ANALYSIS USING GLCM CODE

```
clc;

clear all; close all;

% Load a grayscale image

grayImage = imread('C:\Users\kaviya\Downloads\hd.jpeg');

% Convert to grayscale if needed if size(grayImage, 3) == 3 grayImage =
   rgb2gray(grayImage); end
% Display the grayscale image figure; imshow(grayImage); title('Grayscale Image');
% Compute the GLCM (Gray-Level Co-occurrence Matrix)

% Using an offset of [0 1] for horizontal texture analysis

glcm = graycomatrix(grayImage, 'Offset', [0 1], 'Symmetric', true);

 % Display the GLCM as an image (normalized for visualization) figure; imagesc(glcm);
colormap('gray');
colorbar;
title('Gray-Level Co-occurrence Matrix (GLCM)');
% Compute texture features from the GLCM

stats = graycoprops(glcm, {'contrast', 'correlation', 'energy', 'homogeneity'});
```

% Display the results

fprintf('Contrast: %.4f\n', stats.Contrast); fprintf('Correlation: %.4f\n', stats.Correlation); fprintf('Energy:

%.4f\n', stats.Energy); fprintf('Homogeneity: %.4f\n', stats.Homogeneity);

VIVA QUESTIONS:

1. What is texture analysis?

2. What is the Gray Level Co-occurrence Matrix (GLCM)?

3. Why is GLCM commonly used in texture analysis?

4. What information does GLCM represent?

5. What are the steps involved in creating a GLCM?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:18 DATE:	HYBRID DCT-DWT IMAGE COMPRESSION

AIM:

To write and execute the program for hybrid DCT-DWT image compression.

APPARATUS REQUIRED:

1.Personal Computer

2.Software Tools

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

HYBRID DCT-DWT IMAGE COMPRESSION

```
clc; clear; close all;

I = im2double(imread('peppers.png'));

Igray = rgb2gray(I);

% Step 1: DWT

[LL,LH,HL,HH] = dwt2(Igray,'haar');

% Step 2: Apply DCT on LL band

LL_dct = dct2(LL);

% Step 3: Threshold small DCT coefficients

th = 0.01;

LL_dct(abs(LL_dct) < th) = 0;

% Step 4: Inverse DCT and IDWT

LL_rec = idct2(LL_dct);

I_rec = idwt2(LL_rec, LH, HL, HH, 'haar');

% Step 5: Display

subplot(1,2,1); imshow(Igray); title('Original');

subplot(1,2,2); imshow(I_rec); title('Reconstructed');
```

VIVA QUESTIONS:

1. What is the main objective of hybrid DWT–DCT image compression?

2. Why is DWT used before applying DCT in hybrid compression?

3. What does the LL subband represent in DWT?

4. Which wavelet is commonly used for image compression?

1. How is the compression ratio defined in image compression?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT:

EX.NO:19 DATE:	IMAGE DENOISING AND ENHANCEMENT TECHNIQUES

AIM:

To write and execute the program for image denoising and enhancement technique.

APPARATUS REQUIRED:

1.Personal Computer

2.Software Tools

PROCEDURE:

1. Open the software.
2. Type the program.
3. Save in the current directory.
4. Compile and run the program.

IMAGE DENOISING AND ENHANCEMENT TECHNIQUES

```
clc; clear; close all;
I = im2double(imread('peppers.png'));
Igray = rgb2gray(I);

% Add noise
Inoise = imnoise(Igray, 'salt & pepper', 0.05);

% Denoising using median filter
Ifilt = medfilt2(Inoise, [3 3]);

% Enhancement using histogram equalization
Ienh = histeq(Ifilt);

% Display
subplot(1,3,1); imshow(Inoise);
title('Noisy Image');
subplot(1,3,2);
imshow(Ifilt);
title('Denoised Image');
subplot(1,3,3);
 imshow(Ienh);
title('Enhanced Image');
```

VIVA QUESTIONS:

1. What is the purpose of image denoising?

2. What type of noise is removed using a median filter?

3. What is histogram equalization used for?

4. How does noise affect image quality?

5. What are spatial domain enhancement techniques?

Aim & Algorithm (5M)	Program Execution (10M)	Output & Result (10M)	Viva Voce (5M)	Total (a1) (30M)	Relevance to Po's & PSO's	Timely Submission of Record (b1) (10M)	Grand Total (a1+b1) (40M)

RESULT: